

Element G

After our peers had thoroughly reviewed our decision matrix, we narrowed down our designs between the Omni-Orb and the Flex-Orb, our team held a debate on which of the 2 designs we should target as they had been rated near identically on the matrix, eventually we came to the conclusion that both designs had desirable traits that we could combine into one design we called the Twin-Orb. The Twin-Orb is a design which has an easy to use reloading system, and a complex gravity based puzzle that would take complex thinking skills from the bears to solve for enrichment. The design is composed of two HDPE PVC orbs, an omni-wheel and rotating spoke, the larger orb is fully rotatable and is the main control feature of the orb, the smaller orb is weighted by internal pockets to always be aiming downward so that only when the two orbs align can the food fall out, the smaller orb is able to stay aiming down as the omni-wheel and rotating spoke act as a rudimentary gyroscope. Two major flaws we would come to find with the design were first, after being solved once it becomes significantly easier to solve and loses its enriching attributes, and second, finding a strong enough orb which met our size requirements that also fit in our budget was nigh impossible. We would rework the design with the same original aspects but far more cost efficient and reusable with the spin drum design using two HDPE PVC tubes.

